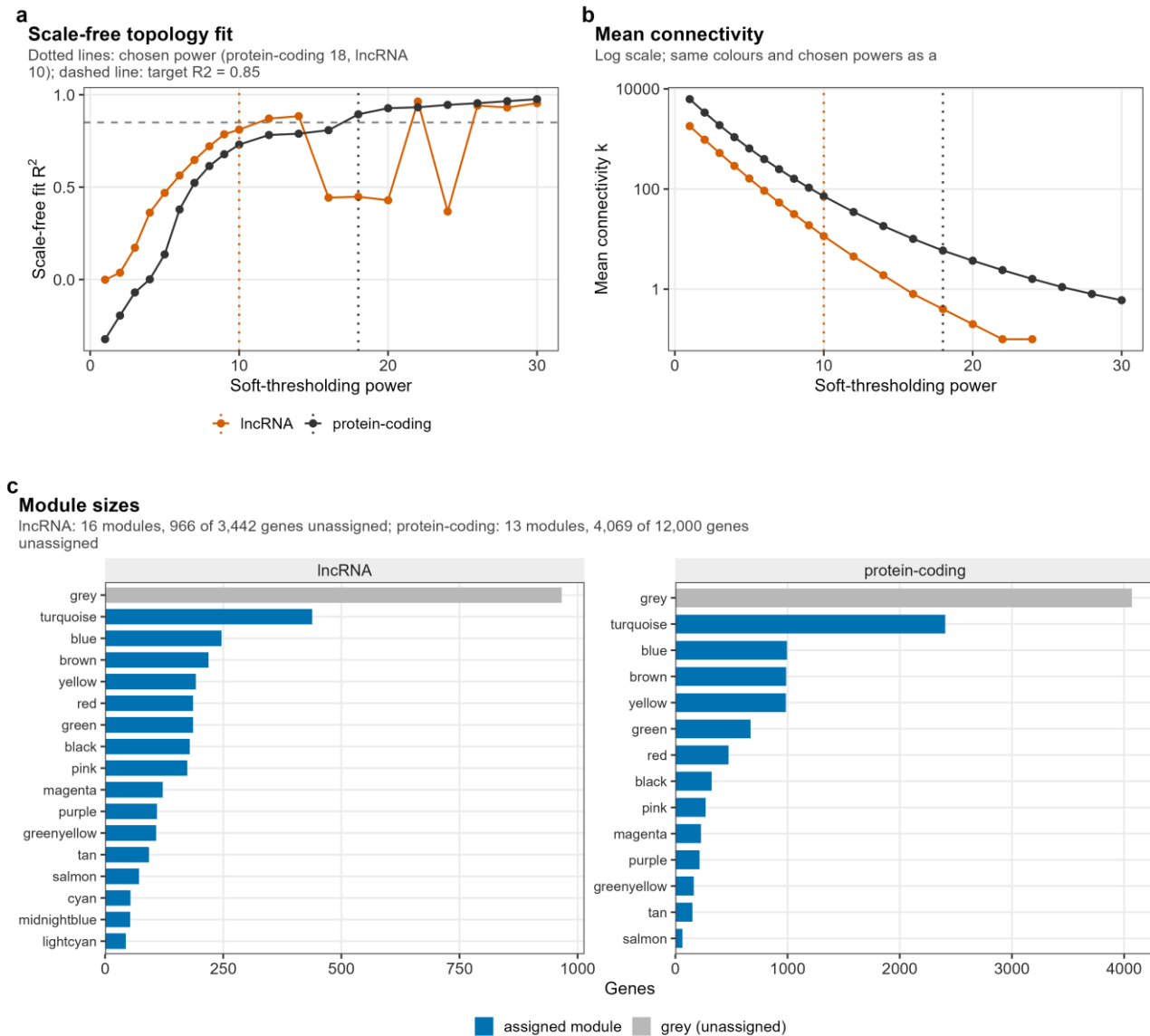
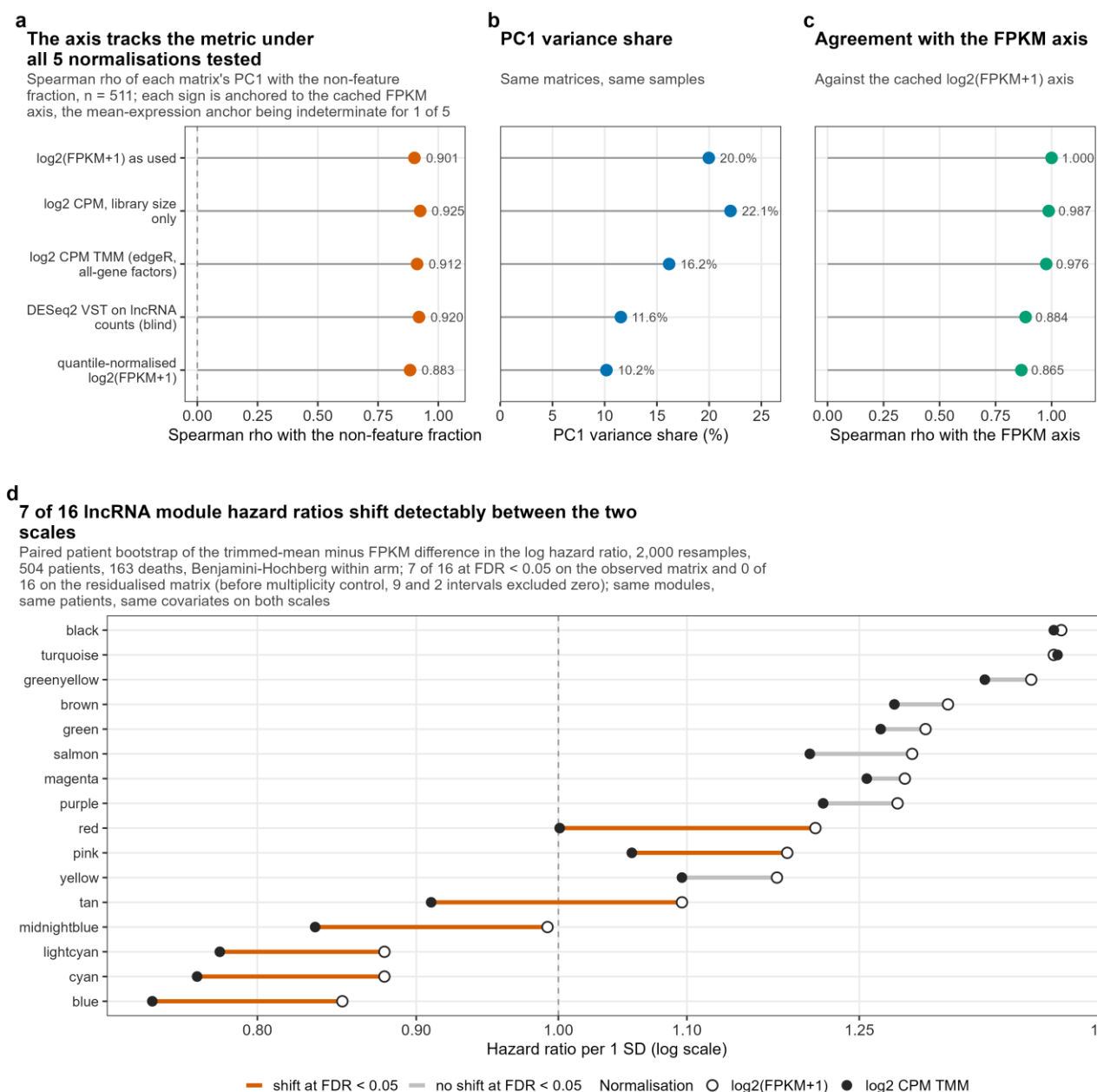




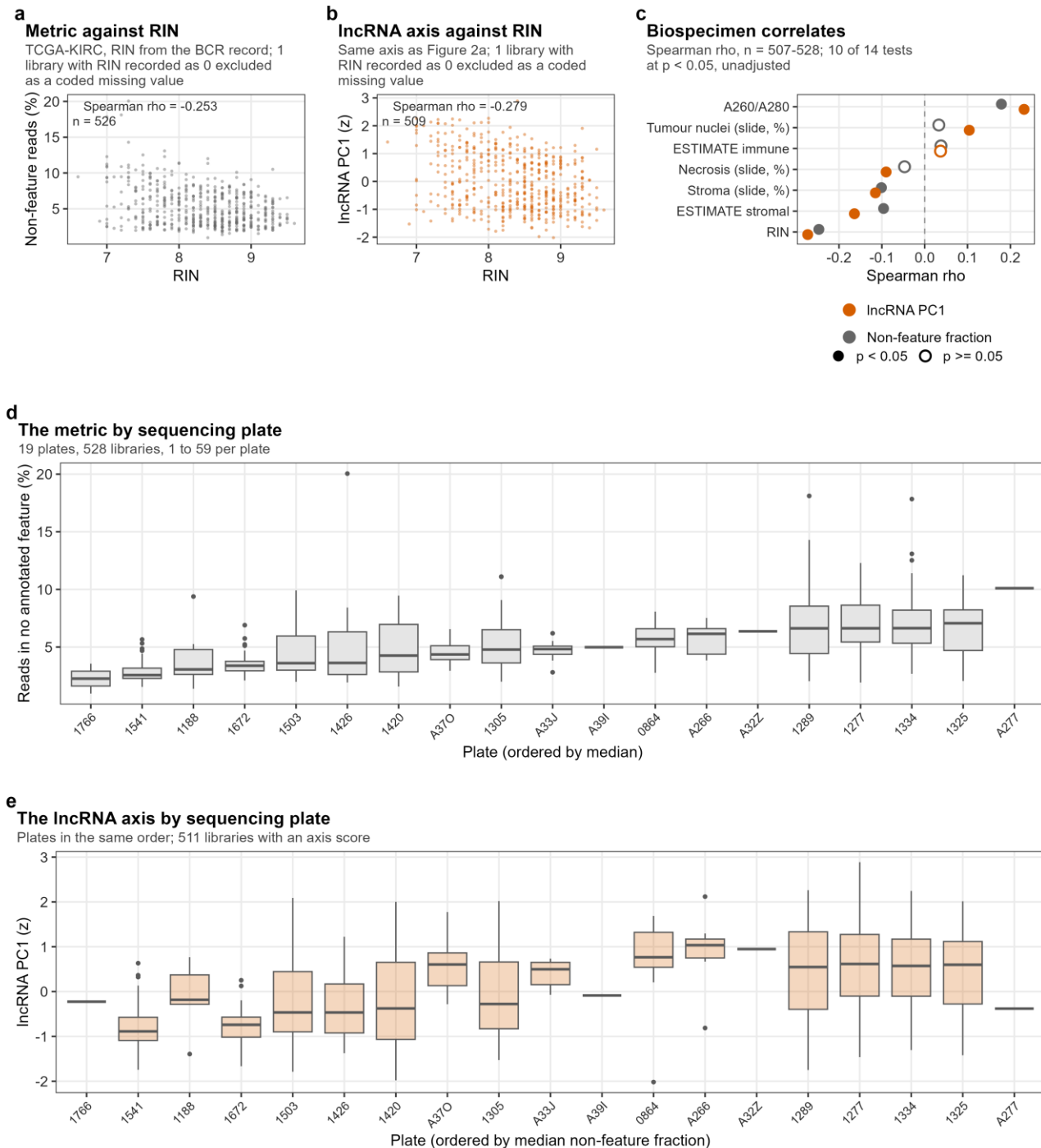
Supplementary Fig. 1 | Network construction diagnostics. **a** Signed scale-free topology fit R -squared against soft-thresholding power for the protein-coding (12,000 genes, 506 samples) and lncRNA (3,442 genes, 511 samples) networks. The horizontal line marks the pre-specified target of 0.85 and the vertical lines the chosen powers, 18 and 10, at which the signed fit is 0.894 and 0.811. The protein-coding power is the lowest reaching the target. The outcome-blind parameter sweep (Methods) fixed the lncRNA power at 10, where the automatic rule would have given 12. **b** Mean connectivity against power on a logarithmic axis, with the same chosen-power lines. **c** Module sizes, with unassigned (grey) genes shown separately. The protein-coding network resolves 13 non-grey modules, leaving 4,069 of 12,000 genes unassigned, and the lncRNA network 16, leaving 966 of 3,442. Source tables: `02_lncRNA_soft_threshold.tsv`, `02_mRNA_soft_threshold.tsv`, `02_network_summary.tsv`, `02_lncRNA_module_sizes.tsv`, `02_mRNA_module_sizes.tsv`.



Supplementary Fig. 2 | The axis under five normalisations. **a** Spearman correlation of the leading lncRNA principal component with the non-feature fraction under five normalisations of the same 3,442 genes and 511 libraries. The five arms are log2(FPKM + 1) as used throughout, log2 counts per million with library size only or with edgeR TMM factors, a blind DESeq2 variance-stabilising transform, and quantile-normalised log2(FPKM + 1). **b** PC1 variance share. **c** Spearman correlation of each arm's PC1 with the cached FPKM axis. Quantile normalisation removes the per-sample mean, leaving that arm's sign anchor indeterminate, so it is quoted as a magnitude. **d** Paired patient bootstrap difference in log hazard ratio between the TMM and FPKM scales for the 16 lncRNA modules, with Benjamini-Hochberg counts in the title and counts before multiplicity control in the subtitle. Source tables: 29_normalisation_pc1.tsv, 29_normalisation_module-contrast.tsv.



Supplementary Fig. 3 | Matched normal-kidney libraries. a Median and interquartile range of the multimapping and non-feature fractions in tumour and matched normal libraries of TCGA-KIRC and CPTAC-3. **b** Projected lncRNA axis (the discovery lncRNA PC1 direction applied to each sample set) against the non-feature fraction in the paired libraries, faceted by cohort and coloured by tissue type. **c** PC1 variance share of each sample set's own lncRNA and protein-coding matrix against a size-matched discovery baseline from 200 resampling draws. The baseline range, the minimum and maximum of those draws, is not a confidence interval. **d** Forest plot of the non-feature fraction, measured on the tumour or the normal library, as exposure for overall survival in the paired patients under three models. Faded rows have an events-per-variable ratio below 5, as does every adjusted model in the panel (71 patients and 26 events in TCGA-KIRC, 129 and 28 in CPTAC-3). Source tables: 25_tumour_vs_normal_distribution.tsv, 25_paired_scores_per_sample.tsv, 25_normal_axis_projection.tsv, 25_normal_metric_survival.tsv.

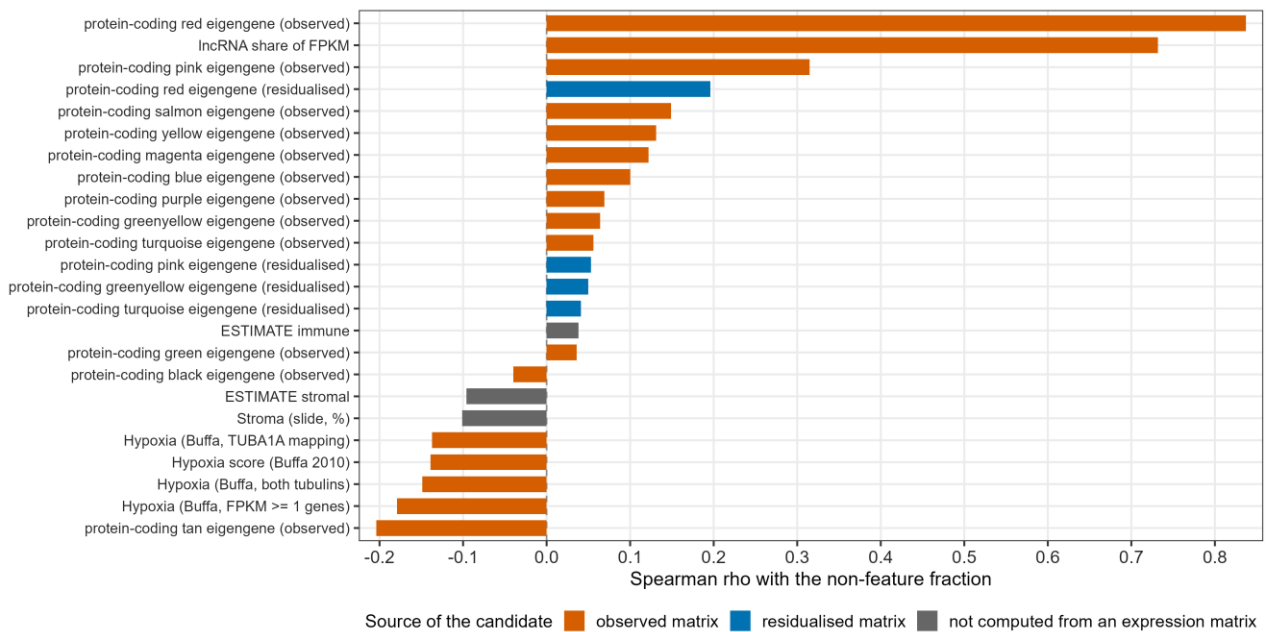


Supplementary Fig. 4 | RNA integrity number and sequencing plate. **a** Non-feature read fraction against RNA integrity number in TCGA-KIRC. One library with an RNA integrity number recorded as 0, 6.6 units below the next lowest, is treated as a coded missing value and excluded from panels **a** and **b**, so their sample size is one below the correlation table's. **b** Observed lncRNA PC1 against RNA integrity number, with the same exclusion. **c** Spearman correlation of the non-feature fraction and the lncRNA axis with seven biospecimen measures: RNA integrity number, A260/A280 ratio, slide percent necrosis, tumour nuclei and stromal cells, and the ESTIMATE stromal and immune scores. **d** Non-feature fraction by sequencing plate, ordered by median. **e** lncRNA axis by plate, in the same order. Biospecimen and batch metadata exist only for TCGA-KIRC. Source tables: 23_biospecimen_kirc.tsv, 23_axis_vs_biospecimen.tsv, 27_axis_scores_per_sample.tsv.

a

What the non-feature fraction correlates with

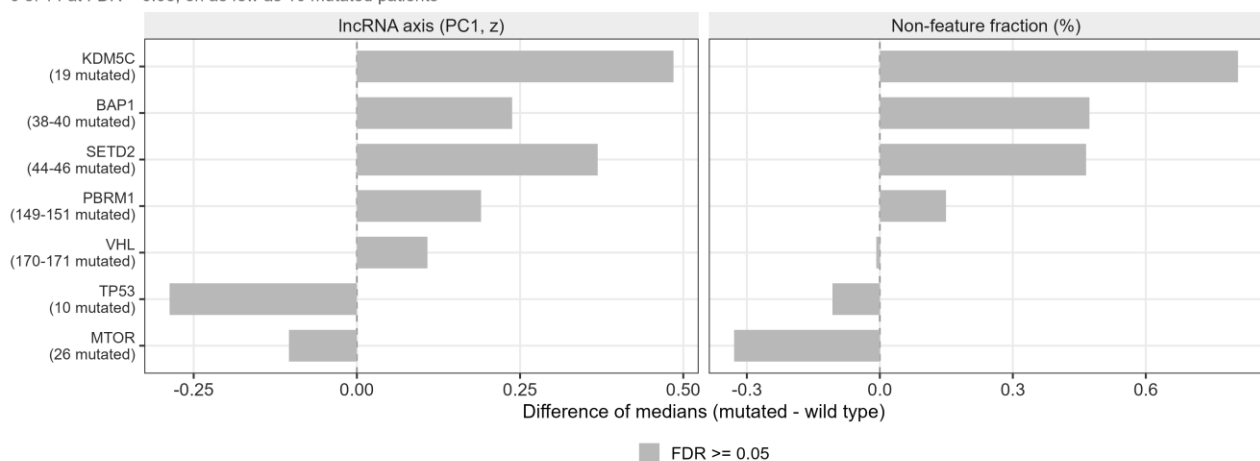
24 strongest of 37 candidates, n = 506-528; observed eigengenes carry the metric by construction



b

No driver mutation is associated with the metric at FDR < 0.05

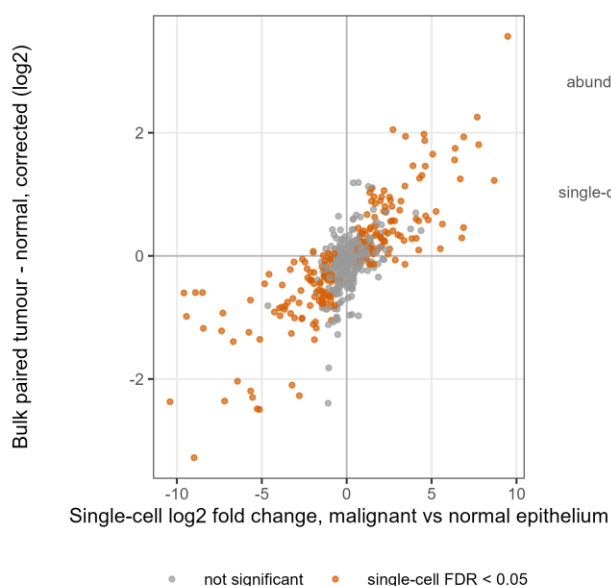
Difference of medians, mutated minus wild type; n = 358-368 patients with a mutation call; 0 of 14 at FDR < 0.05, on as few as 10 mutated patients



Supplementary Fig. 5 | What the non-feature read fraction correlates with. **a** The strongest candidate correlates of the non-feature fraction in TCGA-KIRC, coloured as eigengenes of the observed or the residualised expression matrix, or quantities computed outside it. Observed eigengenes cannot be read as biology, because the metric was measured on the same matrix. The annotated biological candidates are the hypoxia metagene, the proliferation score, the ESTIMATE stromal and immune scores and the slide tumour-nuclei percentage. **b** Difference of medians in the non-feature fraction and the lncRNA axis between mutated and wild-type patients for seven recurrent clear-cell drivers. No test reaches a 5% false discovery rate, as the panel states. Testing is conditional on the 368 of 528 patients with an open-access masked mutation file, whose median non-feature fraction is lower than that of the others. Source tables: 27_metric_biology_correlations.tsv, 28_noFeature_vs_mutations.tsv.

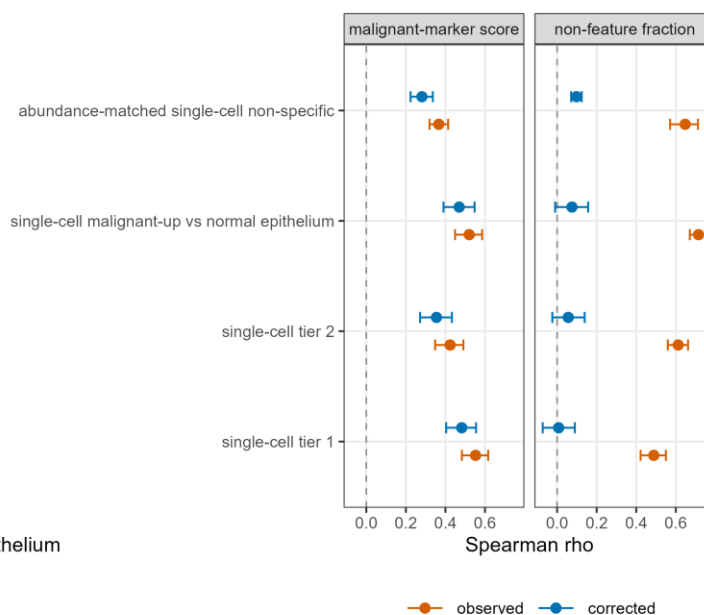
a Tumour-normal differences agree with single cells

507 lncRNAs; Spearman rho 0.775 observed, 0.771 corrected; sign agreement 92.5% and 92.0%



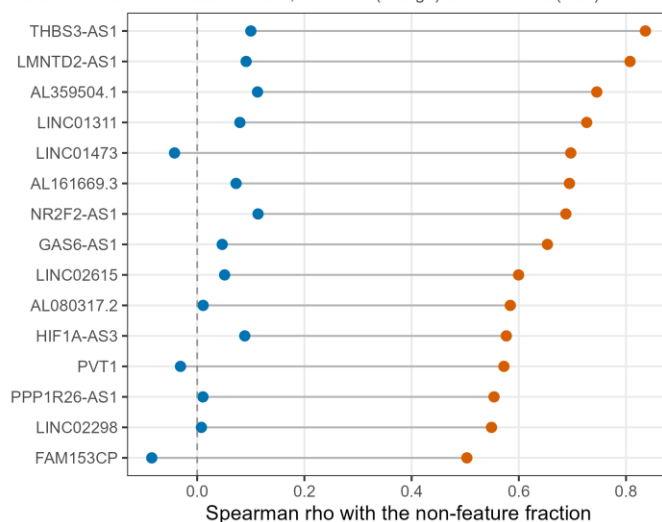
b Correction leaves the tumour signal

Spearman rho of lncRNA scores in 532 tumours; the marker score itself correlates with the metric at 0.22

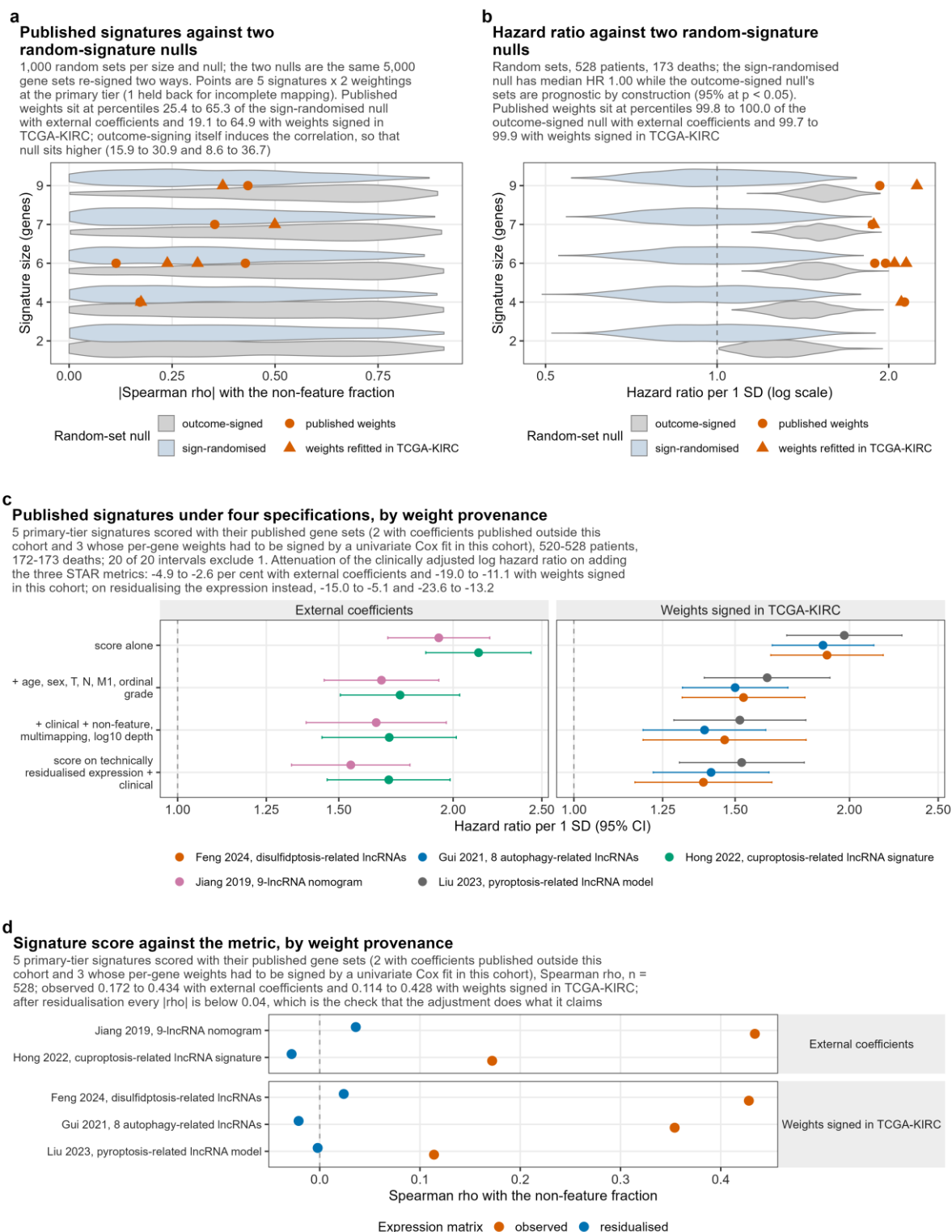


c Candidates set aside as technically sensitive

Correlation with the non-feature fraction, observed (orange) and corrected (blue)



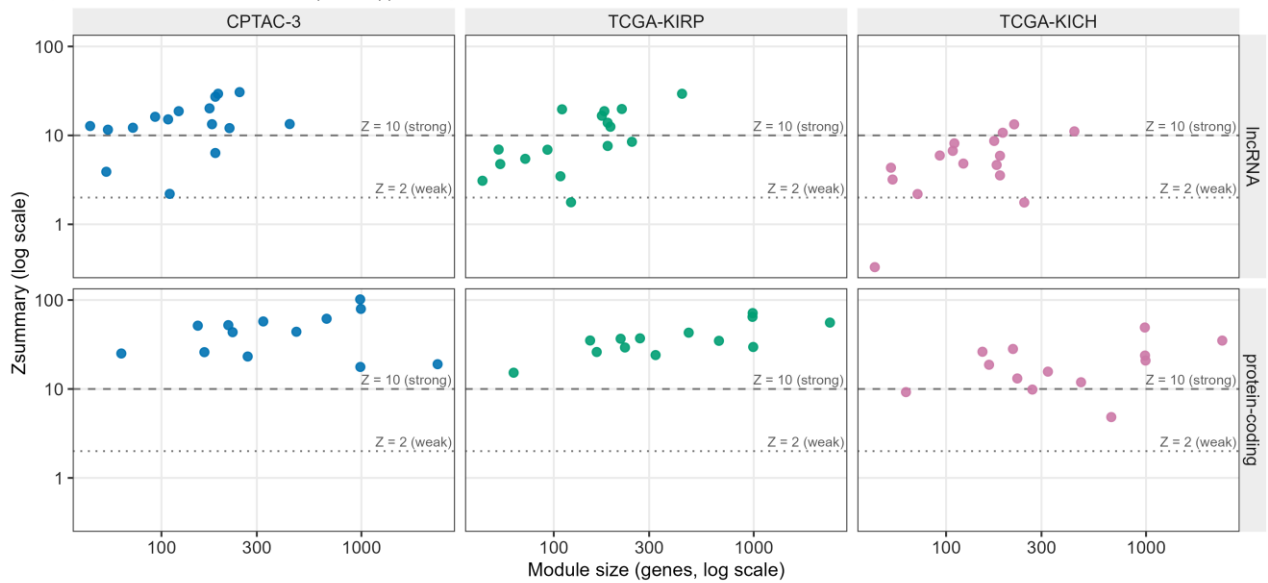
Supplementary Fig. 6 | Single-cell ground truth for the correction. **a** Corrected bulk paired tumour-minus-normal difference against the single-cell fold change of malignant against normal epithelial cells, for 507 lncRNAs in 72 TCGA-KIRC patients. Orange, single-cell false discovery rate below 0.05. **b** Spearman correlation of lncRNA scores in 532 tumours with the malignant-marker score and the non-feature fraction, observed (orange) and corrected (blue), with 95% bootstrap intervals. The null row is 1,000 abundance-matched sets of single-cell non-specific lncRNAs. Correlation with the metric falls by construction, and the informative quantity is marker coherence relative to the null. **c** Correlation with the non-feature fraction, before and after correction, of the 15 single-cell candidates set aside in bulk as technically sensitive. Source tables: 42_sc_truth_concordance.tsv, 42_sc_truth_score_coherence.tsv, 42_sc_truth_technical_candidates.tsv.



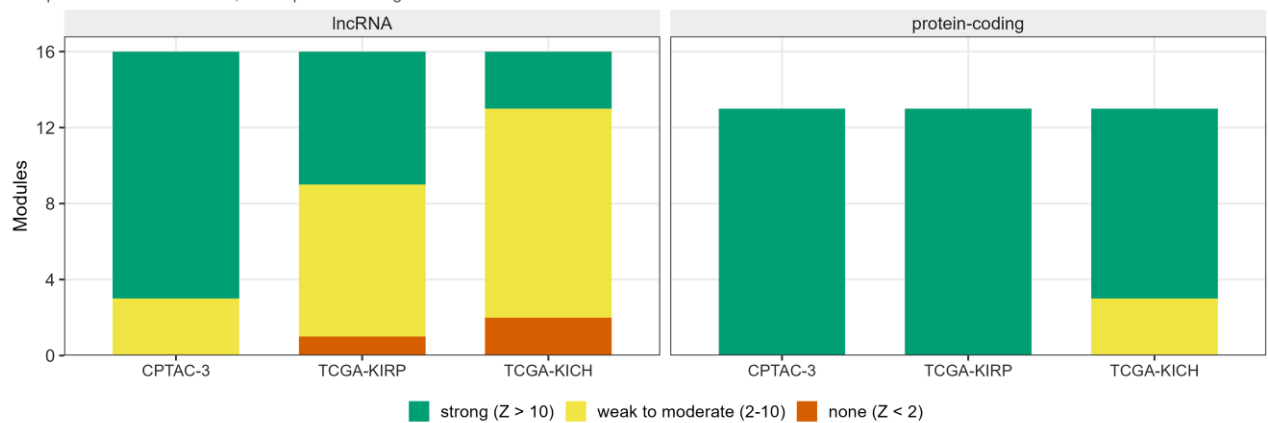
Supplementary Fig. 7 | Published lncRNA signatures against two random-signature nulls. **a** Absolute Spearman correlation with the non-feature fraction (violins) for 1,000 random gene sets at each of five set sizes, under two weightings of the same draws: outcome-signed after Venet and colleagues¹¹, and sign-randomised. The six published signatures are overlaid, with percentiles given separately for the two with published coefficients and the three weighted in sample. **b** Unadjusted hazard ratio per standard deviation (violins) for the same draws and weightings, with the signatures overlaid. **c** Hazard ratio per standard deviation (95% confidence interval) for the five primary-tier signatures under four specifications, grouped by weight provenance, with the two attenuation ranges quoted separately. **d** Spearman correlation of each signature score with the non-feature fraction, observed against residualised expression, by weight provenance. Outcome-signing itself induces correlation with the metric in the null, so outcome-signed percentiles cannot show a signature to be less correlated with the metric than chance. Source tables: 26_random_signature_null.tsv, 26_published_signatures_models.tsv, 26_published_signatures_correlations.tsv, 26_headline_ranges_by_provenance.tsv.

a**Preservation of all 29 modules in each test cohort**

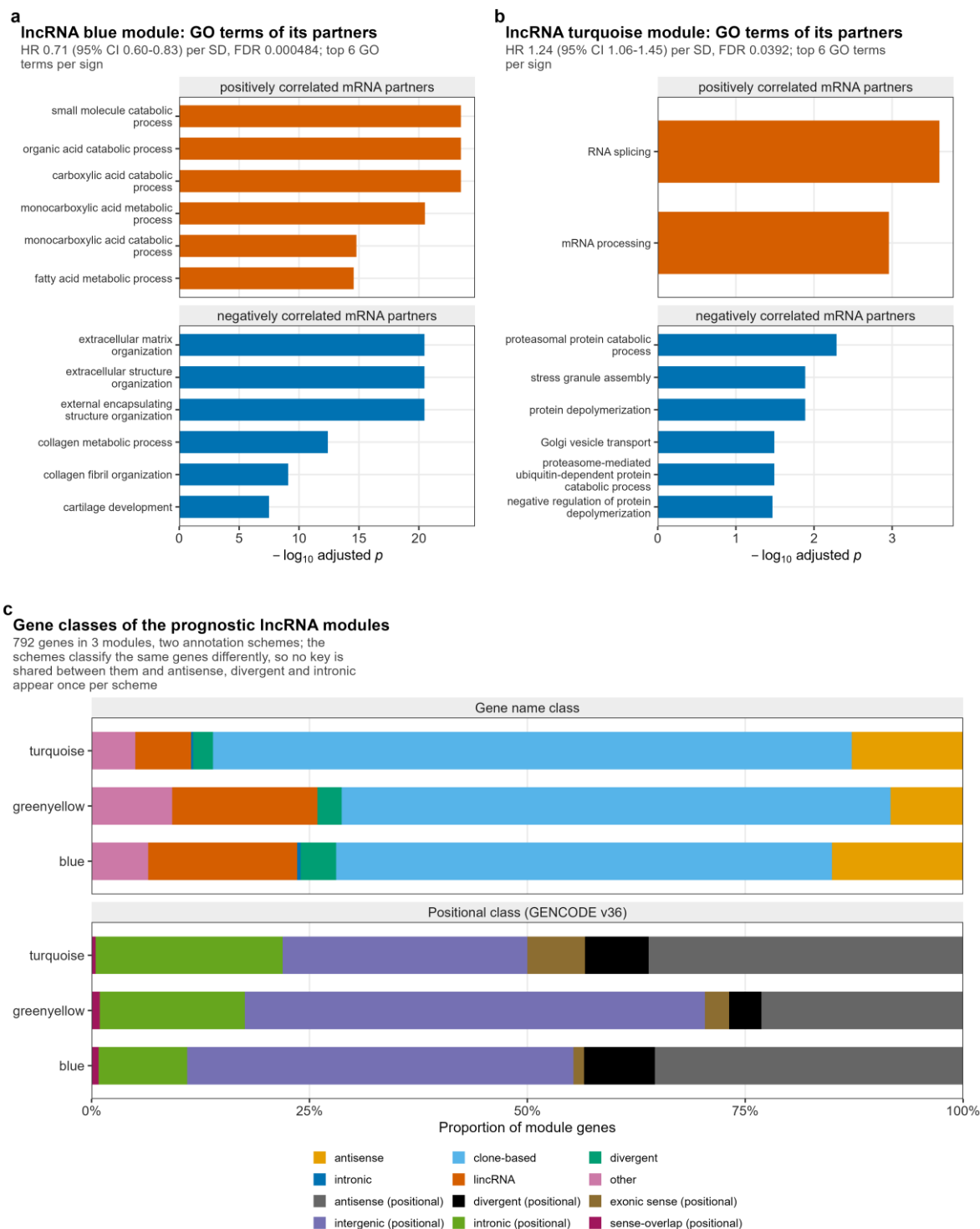
WGCNA modulePreservation; 87 module x cohort pairs (grey and gold excluded); sizes are discovery module sizes, 1 of which modulePreservation reports capped at 1,000

**b****Preservation class counts**

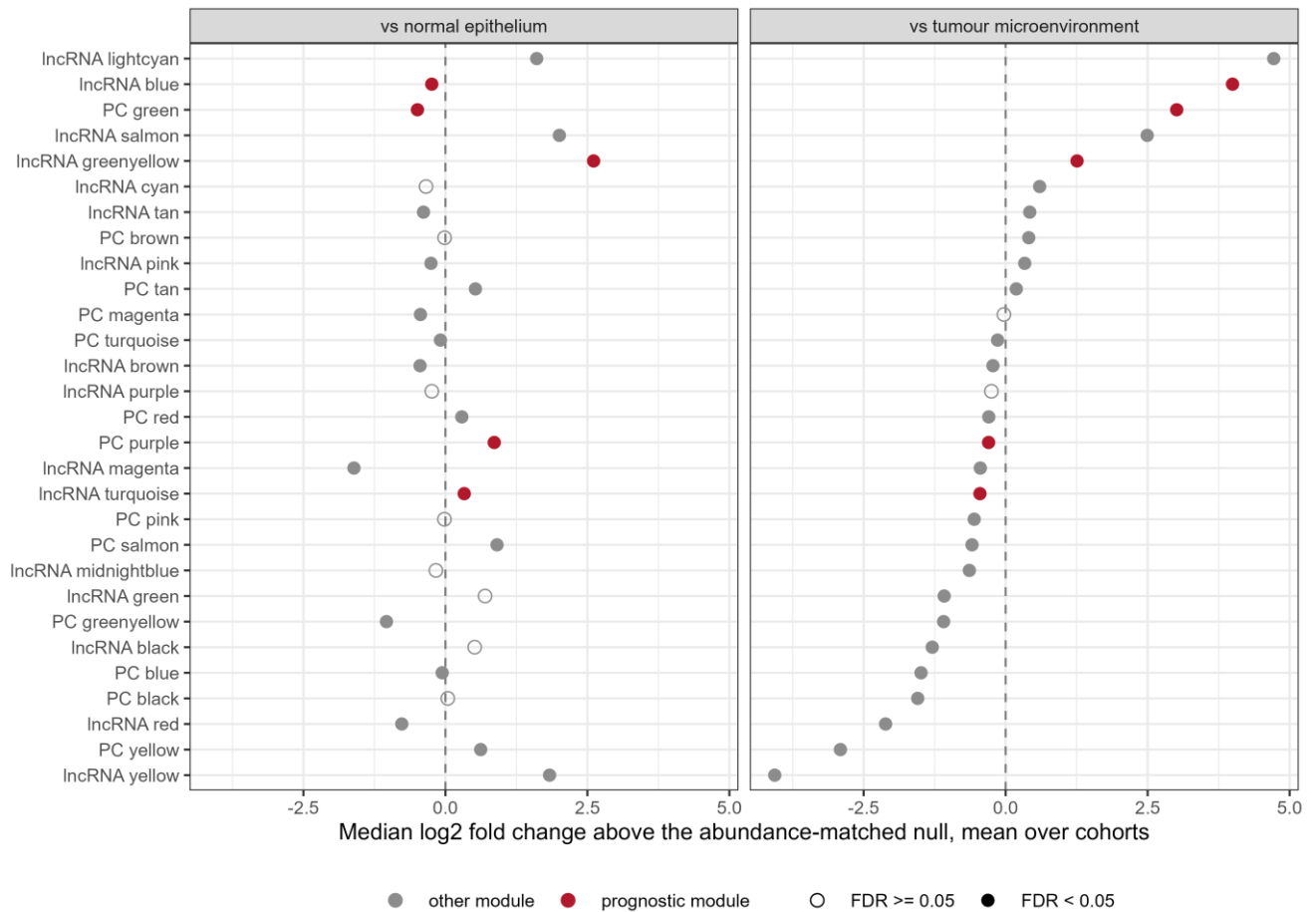
100 permutations for lncRNA, 30 for protein-coding



Supplementary Fig. 8 | Module preservation of the discovery networks in the three external cohorts. a WGCNA Zsummary against discovery module size by network and test cohort, with lines at the conventional bands of Zsummary 2 and 10. The reference is the residualised TCGA-KIRC matrix used to define the modules, and each test network is residualised on its own cohort's STAR metrics. The lncRNA network used 100 permutations, the protein-coding network 30. **b** Stacked counts of modules strongly, weakly to moderately or not preserved, per cohort and network. Strongly preserved lncRNA modules number 13 of 16 in CPTAC-3, 7 of 16 in TCGA-KIRP and 3 of 16 in TCGA-KICH, and protein-coding modules 13 of 13, 13 of 13 and 10 of 13. The gold and grey rows of the underlying tables are WGCNA internal reference sets, not modules. Source tables: 24_module_preservation_lncRNA.tsv, 24_module_preservation_mRNA.tsv, 24_module_preservation_summary.tsv, 02_lncRNA_module_sizes.tsv, 02_mRNA_module_sizes.tsv.



Supplementary Fig. 9 | Partner Gene Ontology terms and gene classes of the prognostic lncRNA modules. **a, b** Gene Ontology biological-process terms of the protein-coding partners of the lncRNA blue (**a**) and turquoise (**b**) modules, drawn as in Fig. 6c for greenyellow. Partners are split by the sign of their correlation with the module eigengene, the 300 most strongly correlated of each sign are tested separately against the 12,000 network protein-coding genes, and up to six terms with the smallest Benjamini-Hochberg adjusted p value are shown per sign. Turquoise has two enriched terms among its positive partners. Because partners are selected by correlation, the adjusted p values rank terms and are not calibrated tests. Each panel title gives the module's hazard ratio under the principal specification. **c** Gene-class composition, as stacked proportions, of the three prognostic lncRNA modules, which hold 246 genes (blue), 108 (greenyellow) and 438 (turquoise), under two annotation schemes. The name-based classes are antisense, clone-based unnamed, divergent, intronic, named long intergenic and other. The positional classes, relative to GENCODE v36 protein-coding genes, are exonic sense, antisense, intronic, sense-overlapping, divergent and intergenic. Source tables: 10_lncRNA_guilt_by_association_GO_signed.tsv (**a, b**), 10_lncRNA_module_composition.tsv and 28_module_composition_by_class.tsv (**c**).

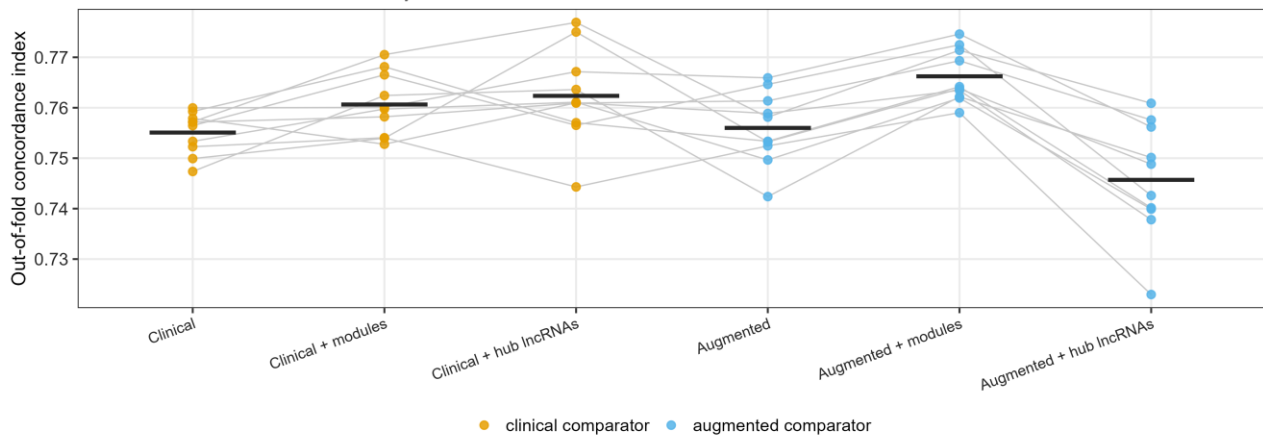


Supplementary Fig. 10 | Single-cell localisation of the 29 discovery modules. For each module, the median log2 fold change of its members minus the mean of 2,000 abundance-matched random gene sets, averaged over cohorts, for malignant cells against normal kidney epithelium (three cohorts, left) and against the rest of the tumour microenvironment (four cohorts, right). Positive values mean higher in malignant cells than a matched random set. Red points, the five prognostic modules. Filled symbols, a within-contrast false discovery rate below 0.05 by Stouffer combination of per-cohort z scores, which overstates precision because genes within a module are correlated. Per-cohort values are in Supplementary Table 19. Source table: 36_sc_module_localisation_combined.tsv.

a

Cross-validated concordance, 10 repeats of 10-fold

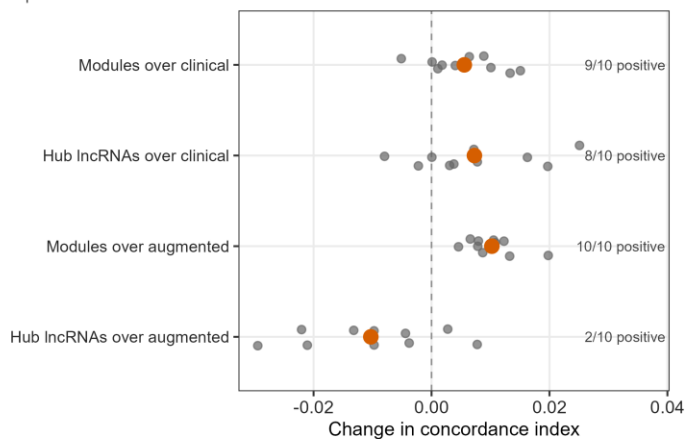
Lines join the same repeat; bars are the mean over repeats. Clinical = age, sex, T, N, M1, ordinal grade; Augmented = clinical + ESTIMATE stromal and immune scores + the three STAR library metrics



b

Increment per cross-validation repeat

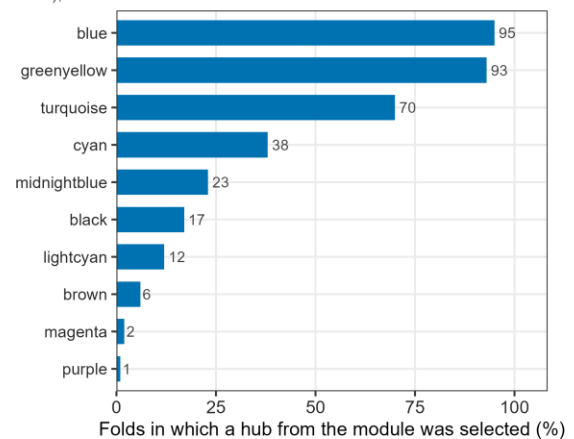
Grey points are repeats, red points the mean over 10 repeats



c

How often each module supplies a selected hub

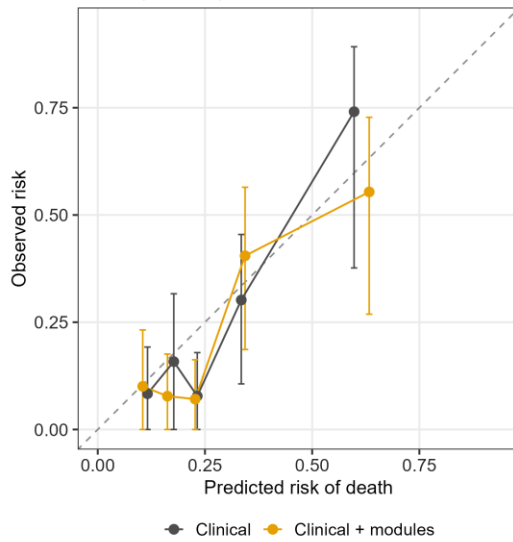
Elastic-net selection inside 100 folds (10 repeats x 10 folds); labels are fold counts



Supplementary Fig. 11 | Cross-validation of hub lncRNAs selected inside each fold. a Out-of-fold concordance index of six models over ten repeats of ten-fold cross-validation in 493 discovery patients with 159 events. The arms are the clinical and augmented models, each alone, with the 29 module eigengenes or with up to 200 hub lncRNAs added under an elastic-net penalty. **b** Per-repeat increment for four comparisons, with the number of repeats in which it is positive. **c** Percentage of the 100 training folds in which a hub lncRNA from each module entered the penalised model, 95 of 100 for lncRNA blue, 93 for greenyellow and 70 for turquoise. Because the candidate pool is chosen inside each training fold, the panel shows the stability of feature selection, not a fixed signature. Source tables: 05_cv_cindex_per_repeat.tsv, 05_hub_module_selection_frequency.tsv.

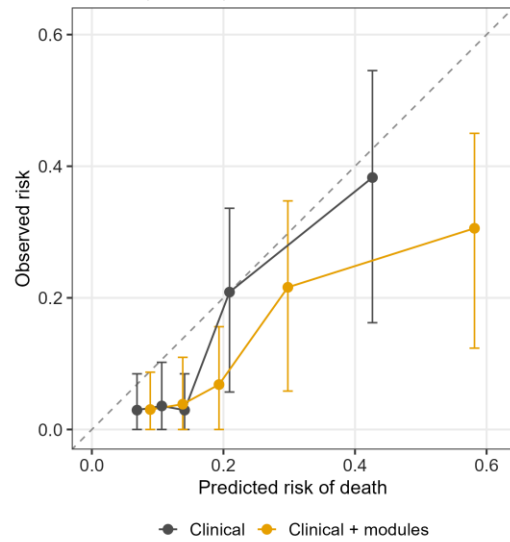
a Calibration at 5 years

CPTAC-3, 23 at risk at 5 years; bars are Kaplan-Meier 95% CIs; slope (95% CI), observed/expected:
Clinical: 0.97 (0.60-1.34), O/E 0.90
+ modules: 0.89 (0.58-1.20), O/E 0.89



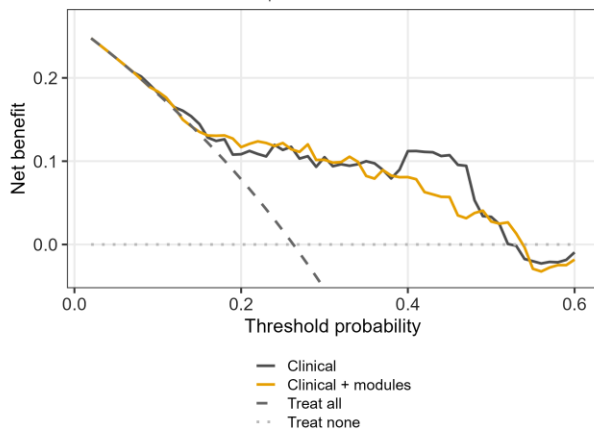
b Calibration at 3 years, discovery-fixed scores

CPTAC-3, 95 at risk at 3 years; bars are Kaplan-Meier 95% CIs; slope (95% CI), observed/expected:
Clinical: 0.97 (0.61-1.34), O/E 0.76
+ modules: 0.89 (0.58-1.20), O/E 0.55



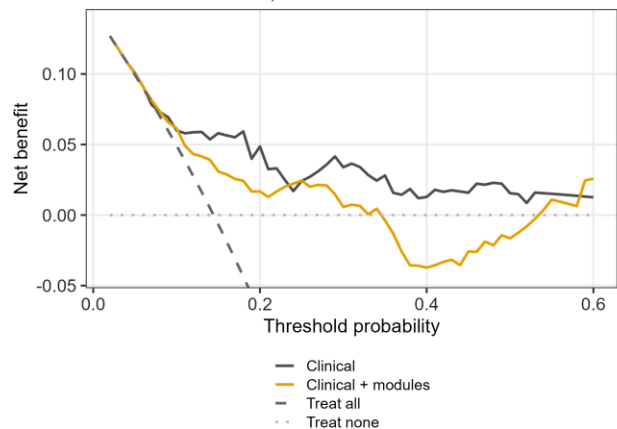
c Decision curve at 5 years, CPTAC-3

Median change in net benefit over thresholds 0.10-0.40: +0.003 (positive at 19 of 31); the treat-all reference runs below the panel

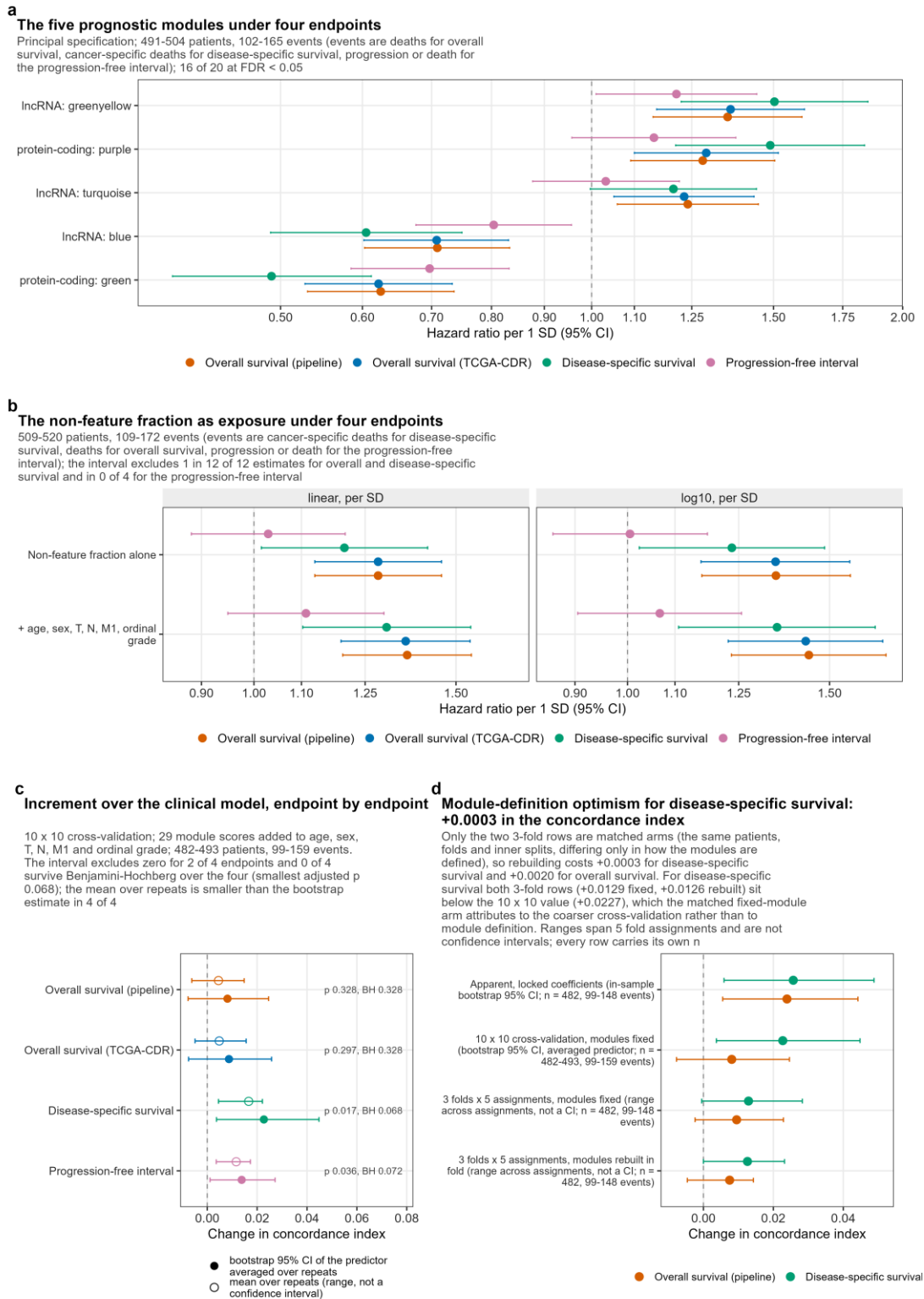


d Decision curve at 3 years, CPTAC-3, discovery-fixed scores

Median change in net benefit over thresholds 0.10-0.40: -0.027 (positive at 3 of 31); the treat-all reference runs below the panel



Supplementary Fig. 12 | Calibration and clinical utility, sensitivity analyses. **a** Calibration at five years in CPTAC-3 under cohort standardisation, by quintile of predicted risk, with the identity line. With twenty-three patients at risk at five years, this is the least stable estimate reported. **b** Calibration at three years under discovery-fixed standardisation, the primary horizon. Under this rule the combined model over-predicts risk about two-fold. **c** Decision curve at five years, cohort standardisation. **d** Decision curve at three years, discovery-fixed standardisation. All decision curves use the baseline hazard fixed in discovery, and none re-estimates it in CPTAC-3. Source tables: 12_calibration_bins.tsv, 12_calibration_summary.tsv, 12_validation_decision_curve.tsv.



Supplementary Fig. 13 | Endpoint sensitivity. **a** The five prognostic modules under four endpoints in TCGA-KIRC (principal specification), with the sample size and event count of each endpoint. **b** The non-feature fraction as exposure under the same endpoints, on linear and log10 scales, with two models per scale. **c** Cross-validated increment in the concordance index from adding the 29 module eigengenes to the clinical model, by endpoint, with Benjamini-Hochberg adjustment over the four endpoints. The mean over the ten repeats is shown beside the bootstrap estimate, the increment of the repeat-averaged out-of-fold predictor, and only the bootstrap estimate carries a confidence interval. **d** Fold-wise network rebuild for disease-specific survival against its matched fixed-module arm, with the ten-by-ten cross-validation row for scale. Only the two three-fold rows are matched arms, as the subtitle states. Source tables: 30_endpoint_modules.tsv, 30_endpoint_exposure.tsv, 30_endpoint_cv_increment.tsv, 30_endpoint_foldwise.tsv.

Supplementary Tables